

WEB SERVICES — ASSIGNMENT 3

Integrate an External SOAP Partner

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1. Context

CampusEats integrates with an external payment gateway to process payments for student food orders. The payment integration uses SOAP while the rest of CampusEats can use REST because the payment partner benefits from a strong WSDL-based contract, structured message-level security, and clearly defined request, response, and fault messages. SOAP is also suitable for enterprise payment integration where reliable and transaction-oriented processing is important. The single partner operation integrated in this assignment is **charge**, which allows CampusEats to submit a payment request and receive a payment result.

Partner details

Business	CampusEats Payment Gateway
Service	Payment Gateway Service
Operation	charge
Endpoint	https://payments.example.com/soap/PaymentGateway

2. HTTP Binding

The SOAP request is carried using an HTTP POST to the endpoint defined by the WSDL service/port. The SOAPAction below is the same value defined in the WSDL binding for the **charge** operation.

```
POST /soap/PaymentGateway HTTP/1.1
Host: payments.example.com
Content-Type: text/xml; charset=utf-8
SOAPAction: "http://campuseats.example.com/payment/charge"
```

```
Endpoint URL:
https://payments.example.com/soap/PaymentGateway
```

3. Discovery

CampusEats can obtain the partner WSDL either from a direct WSDL URL supplied by the payment provider or through a service catalogue/registry. In this design, the catalogue contains the business name, service name, SOAP endpoint, WSDL location, and a tModel-style pointer identifying the technical contract. CampusEats retrieves the WSDL from the listed location and uses it as the contract for its SOAP consumer. No live UDDI server is required for

this assignment.

One-record catalogue / registry entry

Field	Entry
Business	CampusEats Payment Gateway
Service	Payment Gateway Service
Endpoint	https://payments.example.com/soap/PaymentGateway
WSDL	https://payments.example.com/soap/PaymentGateway?wsdl
tModel-style pointer	https://registry.campuseats.example/tmodels/payment-gateway-v1

4. Fault Mapping

The external payment gateway's **card_declined** SOAP fault is translated by the CampusEats payment integration layer into the CampusEats contract error **payment_declined**. Similarly, a partner **gateway_unavailable** fault is mapped to **payment_service_unavailable**. This prevents partner-specific SOAP fault vocabulary from being exposed to CampusEats students and keeps the CampusEats API contract independent of the external provider.

Partner SOAP fault	CampusEats error
card_declined	payment_declined
gateway_unavailable	payment_service_unavailable

Note: This assignment is design-time only. The endpoint and catalogue URLs above are realistic example URLs; no live payment gateway or UDDI server is required.